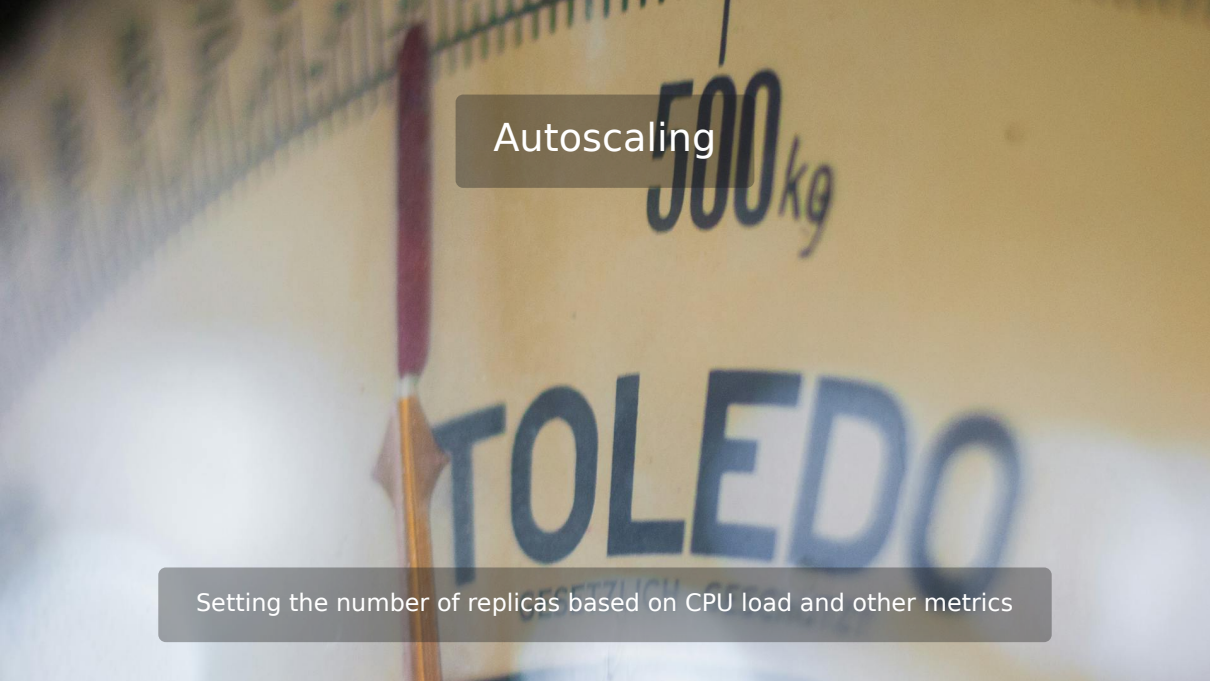
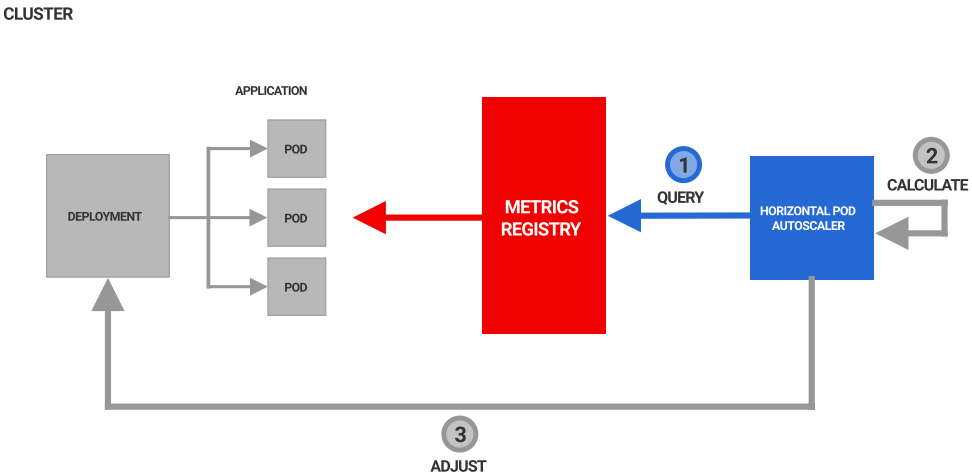


Autoscaling

Setting the number of replicas based on CPU load and other metrics

Overview

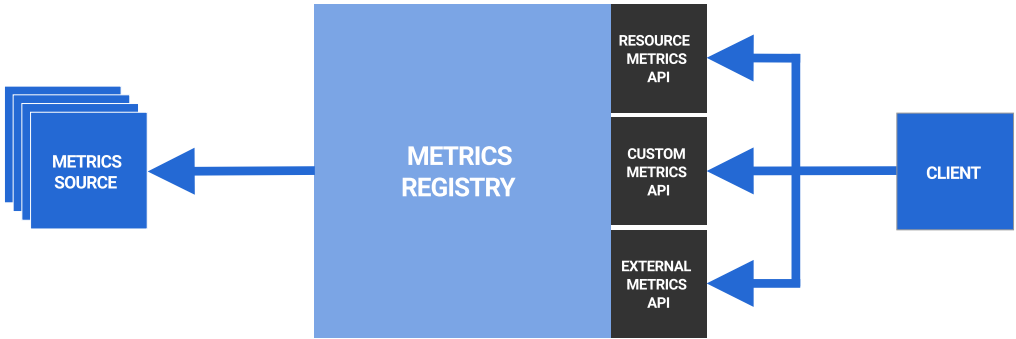
Big picture



Overview

Zooming in on Metrics Registry

- Resource metrics: CPU
- Custom metrics: related to specific pods (e.g. req/s)
- External metrics: not related to a specific pod (e.g. job queue length)



Overview

Example

```yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: hpa-example
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: example
  minReplicas: 3
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
```

Academy content

- Academy course: “Autoscaling in Kubernetes”
- Contains example for custom metrics
- Contains a lab for CPU resource metric

Metrics server

Necessary for having CPU based HPAs

- The CPU HPA depends on having the CPU resource metrics
- To get these metrics you have to have the `metrics-server` installed
- On cloud providers (e.g. Azure, GKE) it's installed by default
- `kubectl top nodes` and `kubectl top pods` also needs this
- You can run these commands to check if you have it or not

Metrics server

Installation on minikube

- The metrics server, but not installed by default on minikube
- https://github.com/kubernetes-sigs/metrics-server#installation
- On minikube, have to disable TLS after installation:
  - `kubectl -n kube-system edit deploy metrics-server`
  - Add `--kubelet-insecure-tls` to args
- You can create CPU based HPAs once `kubectl top pods` is working

Thanks!